WEEK 4 · 23–26 NOVEMBER 2026

Advanced analytics and solution refinement

Two dates, one honest answer — change, uncertainty, harmonisation and scale

Week 4 · Session 1 · practical: 30_advanced_analytics.ipynb

LEARNING OBJECTIVES

What you will be able to do

- Put two acquisitions on a common grid and **co-register** them before comparing pixels
- Map burn severity with dNBR, and confirm a detection with the **full spectrum**
- Separate real change from **artefacts** using an unchanged control area
- Propagate **uncertainty** and say which changes are statistically meaningful
- Harmonise Tanager with Sentinel-2, Landsat and EnMAP, and quantify the difference
- Process a whole scene within a **bounded memory** budget
- Run a refinement loop: sensitivity → error analysis → targeted fix → measure again

1. Two dates on one grid

Geometry before analysis

THE LESSON OF THIS SESSION

Change detection starts with geometry, not with indices.

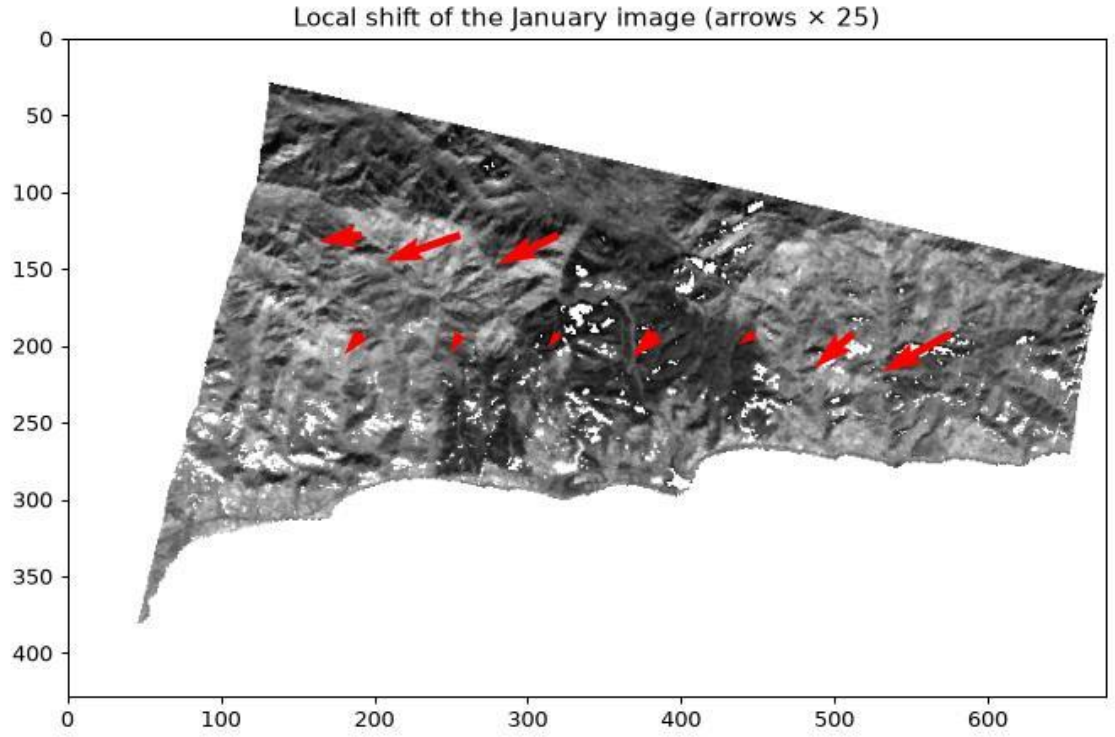
Two orthorectified products of the same sensor over the same place were still about one pixel apart. Subtracting them directly would have produced change along every ridge and edge.

The working example: Malibu, California

- **15 December 2024** — after the Franklin fire, before the January fires
- **23 January 2025** — after the Palisades fire burned the Santa Monica Mountains
- Same sensor, same CRS (EPSG:32611), same 30 m pixel size
- January grid offset from December by **dx = +114 px, dy = −295 px** — a whole-pixel lattice shift
- Overlap: **713 × 677 px = 43 443 ha**, of which 22.7 % is land valid on both dates
  - Because the lattices align, we can slice instead of resample — checked: max difference 0.0

Co-registration with FFT phase correlation

- Cross-correlate in the frequency domain; the peak gives the shift
- Validation first: an **artificial (+3, –5) px** shift was recovered exactly
- Real pair: median **dy = –0.59, dx = –0.82 px** across ten land tiles (≈ 30 m)
- After sub-pixel correction the residual is **dy +0.10, dx –0.27 px**
- NIR correlation between dates improves 0.669 → 0.692



Phase-correlation shift estimates per tile

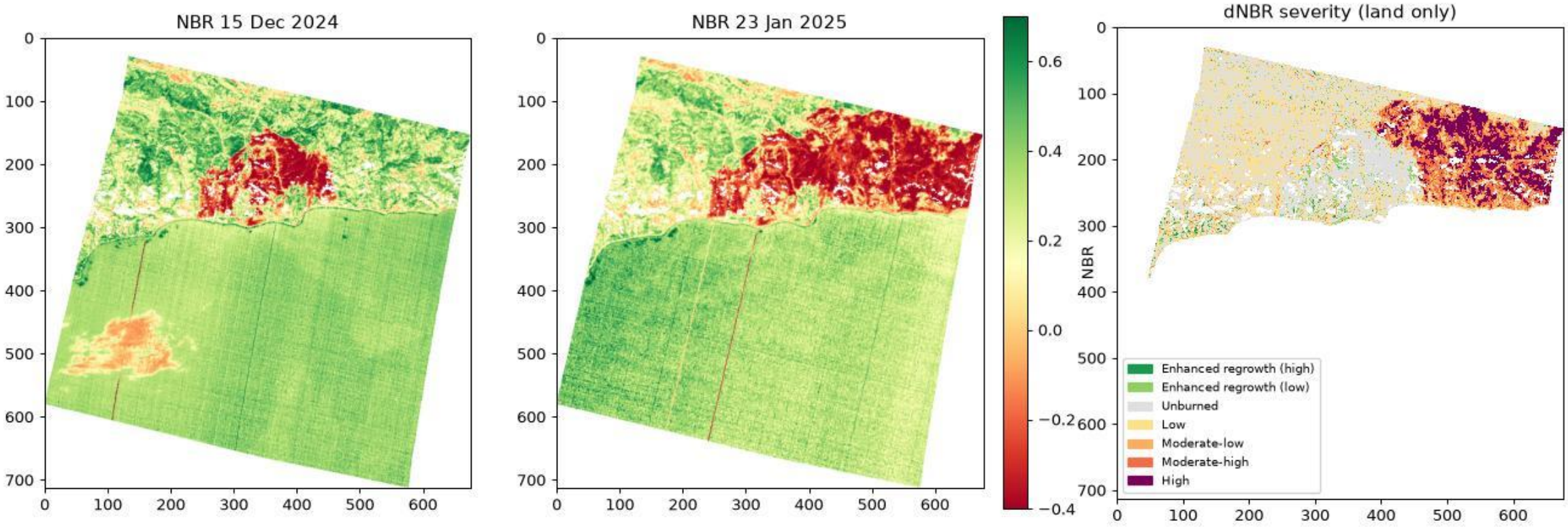
2. Mapping the change

dNBR, severity classes and change vector analysis

NBR and dNBR

- NBR = (NIR − $SWIR_{2200}$) / (NIR + $SWIR_{2200}$) — high for healthy vegetation, low for char and ash
- **dNBR = NBR_pre − NBR_post**: positive where vegetation was lost
- Severity thresholds follow **Key & Benson (2006)**: low 0.10, moderate 0.27, high 0.44, very high 0.66
- dNBR measures **change between your two dates**, not the state of the landscape
  - The Franklin fire burned *before* the December image, so it does not appear in dNBR at all

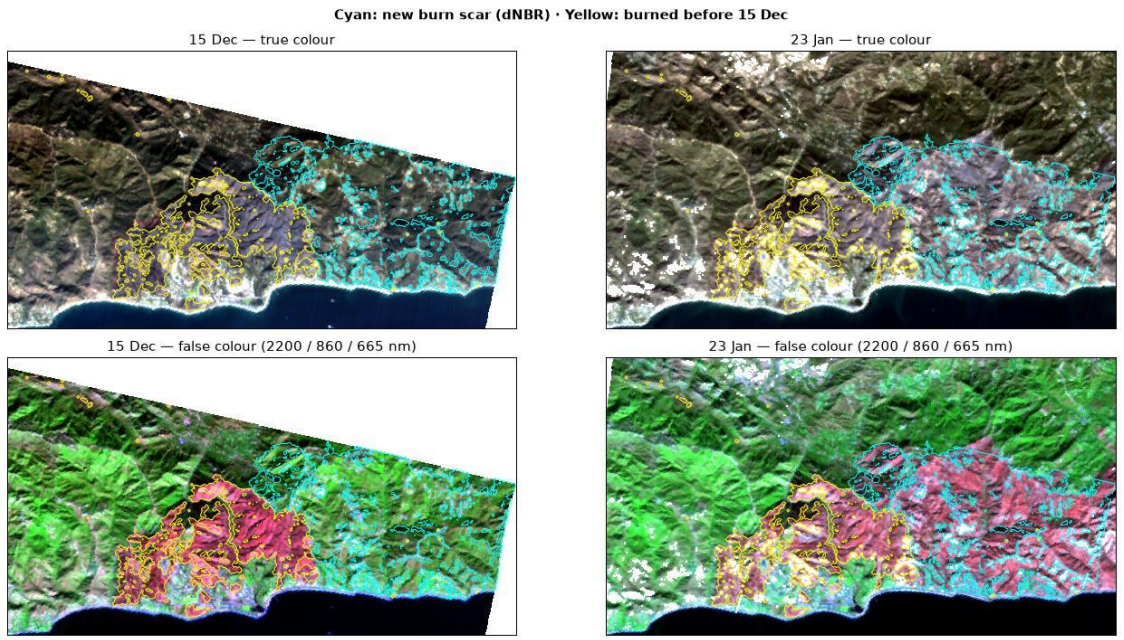
Burn severity across the overlap



dNBR and the USGS severity classes over the Santa Monica Mountains

Is the new scar real? Three independent checks

- **The image:** unburned green in December, dark red in January false colour
- **The shape:** one connected region holds 89 % of all dNBR > 0.27 pixels
- **The spectrum:** the January scar matches the older Franklin scar to **1.7°**
- For contrast: new scar vs stable vegetation is **31.9°** apart
- Area: 28 115 px = **2 530 ha**, centred 118.637° W, 34.061° N



December and January, true and false colour, with both scars outlined

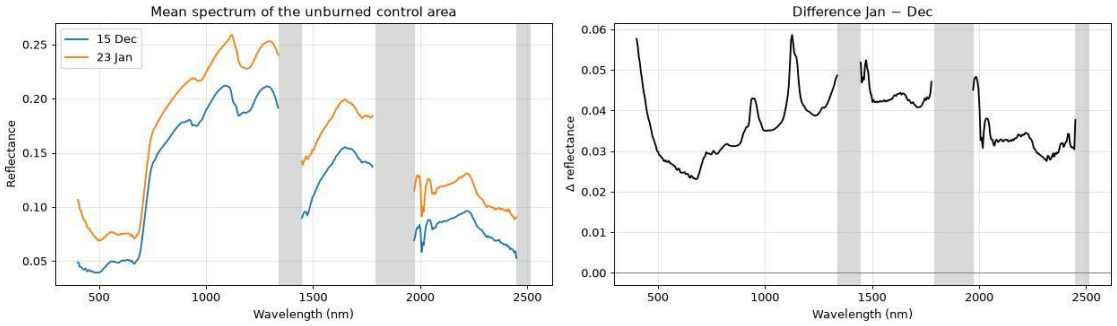
Change vector analysis on full spectra

- Treat each pixel's spectrum as a vector; change is the **difference vector** between dates
- **Magnitude** = how much changed · **direction** = what kind of change
- Burn-like = NIR down and SWIR-2 up; green-up = NIR up and SWIR-2 down
- Inside the new scar: **84.6 %** burn-like · outside it: only **7.0 %**
- Computed tiled over 713 × 677 px × 352 bands × 2 dates in **22.8 s**

3. What the change map gets wrong

Artefacts, control areas and offsets

The January image is systematically brighter



Control-area spectra, December vs January

- Pick a **control area**: land that certainly did not change
- Median dNBR there should be ~0. It is **+0.079**
- That pushes **40 %** of unchanged control pixels into 'low severity'
- 82 % of land outside the scar got brighter in VIS, NIR and SWIR-2 at once
- Cause: viewing geometry (7° vs 20°) and atmospheric correction, not fire

Artefacts and remedies

Artefact	Cause	Remedy
Haze over the ocean	Missed by the cloud mask	Inspect imagery, add a haze test, mask water
Diagonal stripe	Detector column error	Destriping; ignore thin linear features
Cirrus-mask gaps	Conservative flag on bright slopes	Check what the mask removes; relax it
Systematic brightening	Geometry (BRDF) and atmosphere	dNBR offset from a control area
1–2 px misregistration	Ortho over terrain, view angles	Co-registration (section 1)
Earlier burn not detected	It burned before the first image	Choose a true pre-event image

4. How sure are we?

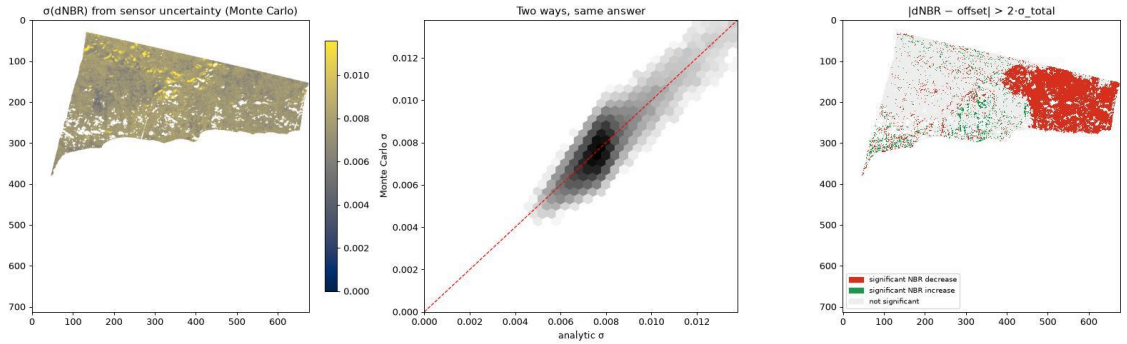
From sensor noise to between-date variability

Two very different uncertainties

- **Sensor uncertainty**: shipped per pixel and band in the Tanager product
- Median relative uncertainty over land: **0.42 % (NIR)**, **1.05 % (SWIR-2)**
- Monte Carlo through the index gives **σ(dNBR) ≈ 0.0075** — analytic propagation agrees (0.0076)
- **Between-date variability** measured on the control area: **σ ≈ 0.087**, more than ten times larger
- Atmosphere, geometry and registration dominate; sensor noise is the small term

What counts as a significant change

- With sensor noise only: **91 % of land** looks 'significant' — and 85.6 % of the control area
- With total σ: **35.4 % of land**, and only **10.3 %** of the control area
- Significant NBR decrease with total σ: **2 941 ha**
- A significance test is only as good as the σ you feed it
- Report the σ you used and where it came from



Monte-Carlo σ(dNBR) and the resulting significance masks

5. Harmonisation and scale

Other sensors, whole scenes

Would another sensor see the same thing?

- Simulate Sentinel-2, Landsat 8/9 and EnMAP by convolving Tanager with their **response functions**
- Compare indices pixel by pixel over 41 577 land pixels
- **EnMAP** is nearly identical to Tanager — similar narrow bands (NBR bias −0.0015)
- **Sentinel-2** NBR carries a bias of **+0.025** (RMSE 0.031); Landsat is similar (+0.026)
- NDVI and NDMI agree much better than NBR — wide SWIR bands are the problem
  - Bandpass differences alone shift indices: harmonise before mixing sensors or thresholds

Same thresholds, different sensors

dNBR threshold	Tanager	Simulated S-2	Pixel agreement
> 0.10	5 215 ha	5 273 ha (+1.1 %)	98.1 %
> 0.27	2 836 ha	2 894 ha (+2.1 %)	99.2 %
> 0.44	2 078 ha	2 192 ha (+5.5 %)	98.8 %

Processing a full scene without running out of memory

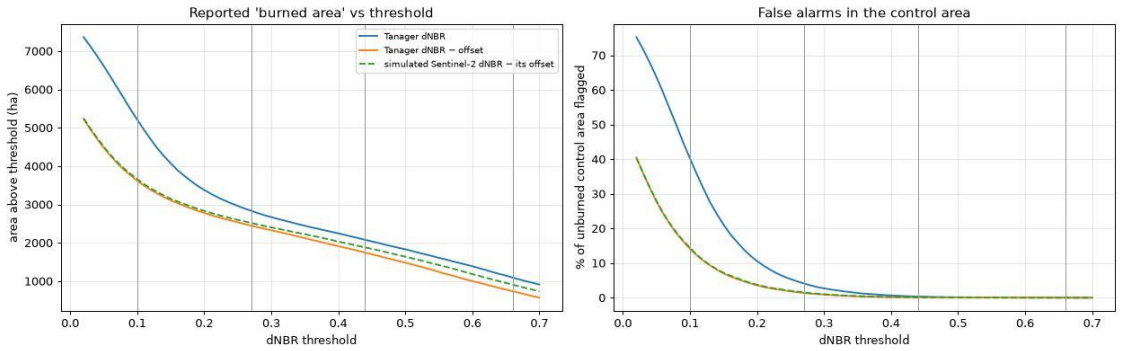
- The full cube (1 063 × 957 px × 426 bands, float32) would need **1.73 GB** in one array
- Tiled processing with a pure per-tile function: **8.3 s**, peak memory **0.24 GB**
- The same pattern as tiled inference in notebook 20 — read, compute, write, discard
- `dask.array` over the HDF5 dataset gives the same answer lazily in **5.6 s**
- Verified identical to the tiled numpy result on all valid pixels

6. Refinement

Sensitivity, error analysis, targeted fixes

How sensitive is your answer to the threshold?

- Mapped area is a steep function of the dNBR threshold
- Raw Tanager: 5 215 ha above 0.10; after the offset correction: **3 621 ha**
- The offset moves the answer more than the sensor choice does
- Simulated Sentinel-2 with its own offset: 3 665 ha — close agreement
- **Report a range and the threshold, never a single bare number**

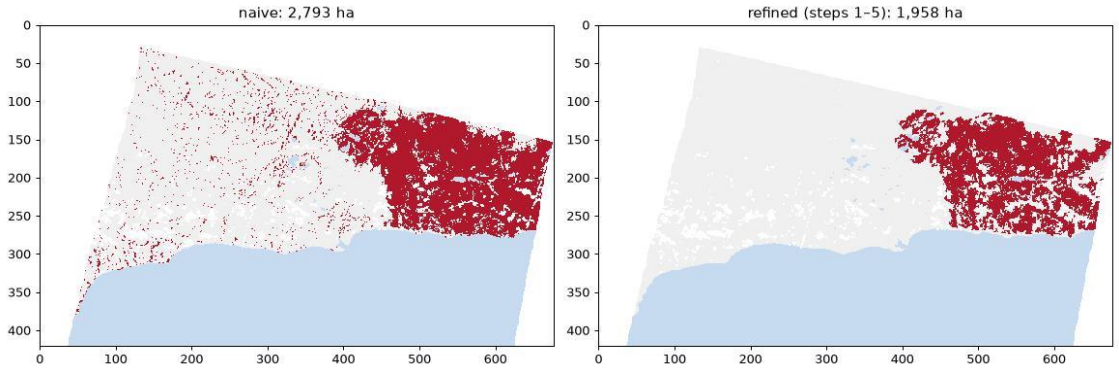


Mapped area against threshold, raw and offset-corrected

Where the false alarms come from

Group	Brighter in all bands	Burn-like direction	Median CVA angle
Control (unchanged) pixels	80.4 %	6.3 %	6.3°
False alarms, dNBR > 0.10	85.6 %	12.9 %	8.4°
False alarms, dNBR > 0.27	83.7 %	16.0 %	15.3°
The real new scar	15.2 %	84.6 %	30.2°

The refinement loop, measured at each step



Naive map versus the refined map after five measured steps

- 0 · naive dNBR > 0.27: 2 793 ha, 1 724 clusters, **4.65 %** control false alarms
- 1 · co-registration: false alarms 4.11 %, scar interior 100 %
- 2 · dNBR offset: **1.37 %** false alarms, 653 clusters
- 4 · require burn-like CVA direction: **0.20 %**, scar 90.1 %
- 5 · minimum mapping unit (10 px): **0.00 %** false alarms, 22 clusters, scar 89.8 %

PRACTICAL

Over to you

▶ `30_advanced_analytics.ipynb` · `150 min`

- Sections 1–2: align the two scenes and co-register them; check the residual shift
- Sections 3–5: dNBR, severity classes, spectral confirmation, change vector analysis
- Sections 6–7: find the artefacts, build a control area, propagate uncertainty
- Sections 8–9: harmonise with other sensors, then process a full scene tiled
- Section 10: sensitivity, error analysis and the refinement loop on your own thresholds

KEY TAKEAWAYS

Recap

- **Geometry first**: CRS, lattice, co-registration better than half a pixel
- **Confirm detections** with the image, the spectrum and independent knowledge
- Measure the **systematic difference on unchanged land** and correct for it
- The honest error bar comes from **between-date variability**, not the sensor's own uncertainty
- **Harmonise** before mixing sensors: bandpass alone shifted NBR by 0.025
- **Tile** everything that needs full spectra; keep the per-tile function pure
- Refinement is a loop: sensitivity → error analysis → targeted fix → **measure again**

Next: Session 2 — Final project implementation and testing (notebook 31)

References and further reading

Key, C.H. & Benson, N.C. (2006). Landscape assessment: sampling and analysis methods. FIREMON, USDA RMRS-GTR-164.

Miller, J.D. & Thode, A.E. (2007). Quantifying burn severity in a heterogeneous landscape with a relative version of dNBR. Remote Sensing of Environment 109, 66–80.

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Olofsson, P. et al. (2014). Good practices for estimating area and assessing accuracy of land change. Remote Sensing of Environment 148, 42–57.

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